

EVERY STEP IS A MOLECULE: VALID-PATH GENERATOR MATCHING FOR CONTROLLABLE MOLECULAR OPTIMIZATION

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ABSTRACT

We formulate source-conditioned molecular optimization as stochastic control of a generator-matched jump process on an operator-closed space of connected, valence-valid molecular graphs. Because every committed state is a complete molecule and every transition is an executable chemical edit, property oracles, structural constraints, and user preferences apply throughout generation, not only at an endpoint. We learn a source-agnostic edit prior by generator matching over this rewrite process, aggregate raw edits at their canonical molecular successor, and steer the fixed-budget embedded jump chain with a finite-horizon Doob transform. The transform reweights existing transition rates, so it cannot leave the legal operator support, and it realizes a specified terminal reweighting exactly when its desirability function is exact. Steering conditions on the remaining edit budget: a preference change made mid-trajectory is served by exact continuation, not restart, and a constraint can be enforced at every intermediate molecule, not only the last. Our primary application is similarity-constrained lead optimization, cast as budgeted multi-objective Pareto exploration; de novo generation from a carbon-tree prior is a secondary demonstration that the operators form a genuine generative process. On enumerable chemical state spaces the transform matches analytic target laws to numerical precision; at scale we study control quality, dynamic preference switching, pathwise constraints, ablations, and robustness to a held-out evaluator. Every step is a molecule, every edge a legal edit, and every preference update changes the controller, not the prior.

1 INTRODUCTION

Lead optimization is a source-conditioned problem. A medicinal chemist begins from a known molecule and searches nearby chemical space for analogues that improve several properties at once—potency, selectivity, solubility, safety, synthetic tractability—while preserving the parts of the molecule responsible for its activity. The search is iterative and revisable: intermediate candidates are evaluated, preferences are revised in light of what is seen, and constraints such as a frozen scaffold or a forbidden toxicophore must hold throughout, not merely at the end. Any state that a property oracle scores, a constraint filters, or a chemist inspects should therefore itself be a complete, chemically admissible molecule. Terminal sanitization cannot make an earlier oracle query meaningful.

Most learned molecular generators do not expose their intermediates as molecules: graph diffusion and flow models evolve node-edge tensors whose intermediate states need not be connected or valence-admissible, and masked or latent generalists denoise a code that is a molecule only at the end (Jin et al., 2018; Lee et al., 2025; Gat et al., 2024). Heuristic graph-edit search does keep every state valid—fragment MCMC, graph genetic algorithms, and reachability-aware editors move between complete molecules (Xie et al., 2021; Jensen, 2019; Qiao et al., 2026)—but it optimizes with hand-designed proposals rather than a learned reference path distribution and steers by endpoint scoring rather than a principled control law. Neither family gives a learned generative *process* whose every state is a molecule *and* whose steering is exact when its value function is. We target that intersection, because it is what budgeted, revisable, constraint-respecting lead optimization needs.

Our framework makes the interface between two established theories explicit for molecular design. Stochastic graph rewriting supplies a state space of complete molecules, a vocabulary of executable edits with state-dependent legality, and continuous-time Markov semantics on the induced state

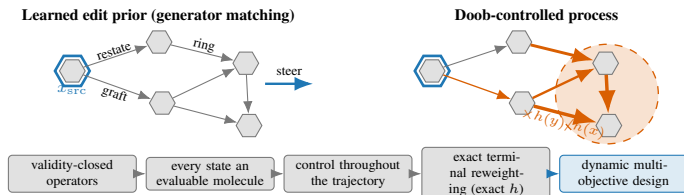


Figure 1: **Valid-path generator matching for controllable molecular optimization.** *Left:* the learned, source-agnostic edit prior is a generator-matched jump process on an operator-closed space of connected, valence-valid molecular graphs; each node is a complete molecule, each edge an executable edit with a learned rate, and a trajectory starts at the source lead x_{src} . *Right:* a finite-horizon Doob transform reweights the *existing* rates by a value ratio $h(y)/h(x)$ (edge thickness, schematic) toward a target region B_z ; multiplying existing rates cannot create edges the prior assigns zero probability, so legal support—and hence validity—is preserved. *Bottom:* the causal chain organizing the paper. Conceptual; no numerical results. Colors stay distinguishable under common color-vision deficiencies and in grayscale (blue = source/control, orange = target/steering, gray = base process).

graph (Danos et al., 2015; Behr et al., 2021). Generator matching supplies a regression principle that recovers a marginal Markov generator from tractable conditional generators (Holderrieth et al., 2025). Composing them yields *valid-path generator matching*: a learned jump process on an operator-closed space of connected, valence-valid molecular graphs, in which probability moves between molecules by executing chemical rules rather than between unconstrained tensors that must later be interpreted or repaired. We call the molecular instantiation COMPOSE. Its causal chain is the organizing idea of this paper (Figure 1): validity-closed operators make every reachable state an evaluable molecule; every state being evaluable makes control applicable throughout a trajectory; a finite-horizon Doob transform turns that into exact terminal reweighting when the desirability is exact; and amortized control turns *that* into dynamic, source-conditioned, multi-objective optimization at scale.

The control result is a finite-horizon Doob transform on the fixed-budget embedded jump chain that editing induces. It reweights the *existing* transition rates by a ratio of value functions, so a controlled trajectory can never reach a state the base prior assigns zero probability: guidance cannot leave the legal operator support, and validity is inherited from the support rather than imposed by the transform. When the value function is exact—as it is on enumerable state spaces—the controlled terminal law is exactly the base law tilted by the desirability, giving exact conditioning on rare substructures, property intervals, and objective-space regions. Because the value function is indexed by the remaining edit budget, a preference change made after inspecting an intermediate is served by an exact continuation transform on the remaining budget rather than by restarting from the lead, and a constraint required at every committed molecule is enforced by restricting the legal support pathwise. These are capabilities that endpoint-conditioned masked generation cannot express and that heuristic edit search cannot make distributionally principled.

We demonstrate the framework primarily on similarity-constrained lead optimization, formulated as budgeted multi-objective Pareto exploration: trajectories start at a source lead, the base edit prior is held fixed, and a goal-conditioned controller reweights the canonical-successor kernel toward under-covered regions of the objective front while a similarity constraint and protected mask keep candidates analogue to the source. The same held-fixed process supports dynamic preference switching, a branching “Pareto fan” of preference-conditioned trajectories from a shared prefix, and hard pathwise constraints. De novo generation from a carbon-tree prior is a secondary demonstration that the operators form a genuine generative process, not the setting where the valid-path property is most consequential; we neither claim nor require that carbon-tree intermediates are useful lead states.

We are explicit about status. The mathematical claims—operator closure, canonical-kernel normalization, the finite-horizon Doob transform with dynamic retargeting, and support preservation—are proved and, on the enumerable benchmark, verified against the implemented process to numerical precision. The large-scale empirical study is in progress (at-scale trace mining, the goal-conditioned controller, the oracles, and the external baselines), so we carry every at-scale value through a result registry and assert no number before its run completes. Concretely: (1) we instantiate generator matching on an executable, operator-closed molecular rewrite system, learning a source-agnostic edit prior whose every state is a complete molecule, and prove operator closure and that aggregating raw

edits at their canonical successor yields a normalized transition kernel on which all controllers act (Sections 3 and 4); (2) we derive a budget-indexed finite-horizon Doob transform of the embedded jump chain, prove it preserves the legal support and realizes the exact terminal tilt under an exact value function, and prove that mid-trajectory preference switching is served by an exact continuation transform distinct from restarting (Sections 5 and 6); and (3) we cast similarity-constrained lead optimization as budgeted Pareto exploration under a goal-conditioned controller and specify its evaluation—a same-base controller comparison, an exact enumerable benchmark, dynamic-steering, pathwise-constraint, ablation, and held-out-oracle studies, with fairness fixed in advance (Sections 7 and 9).

2 PROBLEM SETUP AND NOTATION

State space and edits. A state x is a connected molecular graph over a declared broad-organic vocabulary (elements C, N, O, F, S, P, Cl, Br, I, B, integer Kekulé bond orders, formal charges, implicit hydrogens) in which every atom satisfies its valence constraint. We write $\mathcal{X}_{V,N}$ for such graphs with at most N heavy atoms; it is a *bounded, discrete, combinatorial* state space, not a manifold. The representation excludes stereochemistry, isotopes, and radicals (Section 11), and states may carry formal charges but no edit changes them, so net charge and charged centers are preserved. At x , the finite *legal-mark* set $\mathcal{A}(x)$ collects the executable edits admissible at x ; a mark $a = (\text{family}, \text{operands})$ names a typed rewrite at a match, and an executor applies it, $T(x, a) = y$, with T mapping $\mathcal{X}_{V,N}$ into itself (Proposition 1), so the process never visits an invalid intermediate. Marks come from 8 production families (atom_insert, atom_delete, atom_restate, bond_reorder, bond_reroute (graft), ring_system_grow, ring_system_delete, ring_system_restate; registry in Appendix B).

Trajectories, objectives, constraints. For optimization we initialize $x_0 = x_{\text{src}} \in \mathcal{X}_{V,N}$ and take a fixed K edit steps (the budget); b is the budget remaining. Editing thus uses the fixed- K embedded jump chain and discards the continuous-time hazard (Section 4). The base prior is source-agnostic—it reads only the current molecule and a clock, never x_{src} ; source memory enters only through initialization, a protected mask m (atoms/bonds that may not change), a similarity constraint $\text{sim}(x, x_{\text{src}}) \geq \tau$, and the controller. A property map $F : \mathcal{X}_{V,N} \rightarrow \mathbb{R}^d$ gives the d objectives; we keep an empirical nondominated archive $\mathcal{A}$ and a feasible region $\mathcal{F}$, and compute hypervolume over $\mathcal{A} \cap \mathcal{F}$ against a fixed reference point (Section 10).

Preferences and desirability. A goal descriptor z encodes the current preference—a weight vector, an objective-space region $\mathcal{B}_z$, or a reference direction—and the controller steers toward a positive terminal desirability, in one of two forms:

$$(\text{soft}) g_z(x) = \exp\{\beta U_z(x) - \lambda d(x, x_{\text{src}})\} \mathbf{1}[c(x)], \quad (\text{hard}) g_z(x) = \mathbf{1}[F(x) \in \mathcal{B}_z \wedge \text{sim}(x, x_{\text{src}}) \geq \tau], \quad (1)$$

with U_z a preference utility (Chebyshev or box), $d(\cdot, x_{\text{src}})$ penalizing departure from the source, and $\mathbf{1}[c(x)]$ (1 the indicator) any state-level predicate. The soft form realizes a Boltzmann tilt; the hard form realizes exact conditioning on a reachable target event (Section 5). The base prior and operator system are trained artifacts; the value function realizing g_z at scale is the controller of Section 6, exact where enumerable and evaluated as an approximation otherwise. “Universal” means one edit prior shared across sources and objectives *within the declared vocabulary*, not across all chemical space.

3 VALID-PATH GENERATOR MATCHING

Operators and legal support. Each family is a typed graph rewrite with an application condition: *atom insert/delete* grow or trim a peripheral atom (carrying the incident bond order); *atom restate* changes an atom’s valence state within the vocabulary; *bond reorder* changes a non-ring bond order; *bond reroute* (Graft) detaches a bridge-attached pendant fragment and reattaches it atomically; and the *ring-system* family grows, deletes, or restates a whole supported ring system (mono-/fused/bridged/spiro, aromatic or saturated). A mark instantiates a family at a match, and $\mathcal{A}(x)$ is the set of marks whose condition holds and whose executed successor is a valid, connected molecule distinct from x (self-loops excluded). Two legacy families are disabled everywhere (cycle_insert, cycle_attach); the registry with inverses is Appendix B.

Proposition 1 (Operator closure). *For $x \in \mathcal{X}_{V,N}$ and $a \in \mathcal{A}(x)$, assuming (i) the heavy-atom count stays within N , (ii) each family’s rewrite preserves the valence constraint at every atom it touches and leaves untouched atoms unchanged, and (iii) the successor is required connected, we have $T(x, a) \in \mathcal{X}_{V,N}$. Hence every reachable state is a connected, valence-valid molecule.*

Proof sketch. $\mathcal{A}(x)$ is the fiber of marks passing the executor’s guards (validate source; check condition; execute; re-validate successor for per-atom valence, connectivity, and sanitizability). It suffices that each family preserves the invariants at the atoms it edits, verified family by family in Appendix A.2; closure is thus a property of the operators and the legality filter, not inferred from sampled validity. $\square$

Generator-matching parameterization. The learned process is a continuous-time Markov jump process whose marked off-diagonal rate factors into a state rate and a conditional mark law,

$$q_\theta(x \rightarrow a, t) = \Lambda_\theta(x, t) p_\theta(a \mid x, t), \quad \Lambda_\theta(x, t) = \text{softplus}(s_\theta(x, t)) \cdot \mathbf{1}[\mathcal{A}(x) \neq \emptyset], \quad (2)$$

so $\Lambda_\theta \geq 0$ vanishes at states with no legal mark (a terminal, no numerical fallback), and $p_\theta(\cdot \mid x, t)$ is a categorical over *complete legal marks*, factorized as family times within-family action. Illegal marks get zero probability and, since p_θ normalizes, $\sum_a \Lambda_\theta p_\theta(a \mid x) = \Lambda_\theta$: the total hazard is the productive exit rate, and training and sampling share this factorization. Given a supervised transition with target mark a^* and rate r , generator matching regresses the model marked rate onto the teacher’s via the Poisson (generalized-KL) Bregman divergence; up to target-only constants,

$$\ell_\theta(x, t) = \Lambda_\theta(x, t) - r[\log \Lambda_\theta(x, t) + \log p_\theta(a^* \mid x, t)], \quad r > 0, \quad (3)$$

and $\ell_\theta = \Lambda_\theta$ at a terminal ($r=0$), driving $\Lambda_\theta \rightarrow 0$. An illegal or out-of-mask teacher mark yields $\log p_\theta = -\infty$ and a non-finite loss—a deliberate guard forcing every teacher mark into the factorized fiber the model normalizes over (Appendix A.3).

Source-agnostic prior; source-conditioned control. The base process (2) observes only (x, t) ; the immutable x_{src} enters generation only through $x_0 = x_{\text{src}}$, the source-relative desirability (1), the protected mask, the similarity constraint, and the controller (Section 6). This keeps the molecular dynamics reusable across sources and objectives, keeps the Doob analysis clean (one process, different boundary conditions), and makes a preference change a controller swap that leaves the prior fixed.

Editing training distribution. The prior is trained on real molecule-to-molecule trajectories from three layers so it learns transformations, not identity reconstruction: compiled one-cut matched-molecular-pair analogue paths; sparse shared-scaffold analogue paths (nearest analogue neighbours within a Bemis–Murcko group, never all-pairs cliques); and synthetic reversible legal-corruption paths, all taught both directions. A hierarchical sampler draws a layer (proposed shares $\sim 50\%/\sim 35\%/\sim 15\%$, an ablation target finalized from at-scale statistics), then a *measured* path-length curriculum bin (micro / lead-opt / scaffold, never preset), then a balanced example under anti-concentration safeguards that drop identity traces and cap per scaffold, transformation, and family, each with a signal and a gate. The corpus scope, census, mining, safeguards, and pilot characterization are Appendix D.

4 THE CANONICAL MOLECULAR SUCCESSOR KERNEL

Several distinct marks can name the same chemical transition—symmetric matches, alternative executable lowerings, or different slot presentations of one relabeled graph. Control must act on molecules, so we aggregate marks at their canonical successor before steering. For $x \in \mathcal{X}_{V,N}$ and successor y ,

$$\mathcal{G}_y(x) = \{a \in \mathcal{A}(x) : T(x, a) \cong y\}, \quad P_\theta(y \mid x, t) = \sum_{a \in \mathcal{G}_y(x)} p_\theta(a \mid x, t), \quad (4)$$

with $\cong$ isomorphism on canonicalized molecules; P_θ is the transition kernel of the embedded jump chain over molecular states.

Proposition 2 (Canonical-kernel normalization). *The groups $\{\mathcal{G}_y(x)\}_y$ partition $\mathcal{A}(x)$ (each legal mark has one canonical successor; self-loops excluded), so $P_\theta(\cdot \mid x, t)$ is a distribution over distinct successors: $\sum_y P_\theta(y \mid x, t) = \sum_{a \in \mathcal{A}(x)} p_\theta(a \mid x, t) = 1$.*

Proof sketch. Isomorphism is an equivalence on $\{T(x, a)\}$, and $a \mapsto [T(x, a)]$ assigns each mark a unique class, so the fibers are disjoint and cover $\mathcal{A}(x)$; summing the normalized mark law over a partition preserves mass one. The successor-level (group-logsumexp) supervision of Graft, and why it does not over-count symmetric non-Graft successors, is Appendix A.4. $\square$

Three distinct objects share these rates and must not be conflated. (i) *Mark training* supervises $p_\theta(a \mid x)$ via (3), with Graft scored at the successor group so generator matching supervises the molecular transition. (ii) *Successor sampling* draws $y \sim P_\theta(\cdot \mid x)$ from (4) and keeps a representative

executable mark; a canonical-successor quotient sampler induces exactly this kernel, verified equal to the raw mark sampler’s to within 10^{-5} across five state classes (Appendix E). (iii) *The timed process*: de novo generation is a timed CTMC whose hazard sets the waiting time (Gillespie), whereas editing *discards* the hazard, taking a fixed budget of steps from P_θ and reading only whether a state is terminal. Editing is thus the embedded jump chain over a fixed budget; all controllers act on this one kernel P_θ , and we reserve “CTMC”/“Gillespie” for de novo.

5 FINITE-HORIZON DOOB CONTROL

Editing is a fixed-budget embedded jump chain with (possibly step-dependent) kernels $P_0, \dots, P_{K-1}$, $P_k \equiv P_\theta(\cdot \mid \cdot, t_k)$. Because the hazard is discarded, the exact control object is the finite-horizon Doob transform of this chain, whose value is indexed by the step—equivalently the remaining budget $b = K - k$ —which is what the controller conditions on.

Theorem 3 (Finite-horizon Doob transform). *For a terminal desirability $g : \mathcal{X}_{V,N} \rightarrow \mathbb{R}_{\geq 0}$ define $h_K(x) = g(x)$ and $h_k(x) = \sum_y P_k(x, y) h_{k+1}(y)$ for $0 \leq k < K$, and on $\{h_k(x) > 0\}$ the controlled kernel $P_k^g(x, y) = P_k(x, y) h_{k+1}(y) / h_k(x)$. Then (i) $P_k^g(x, \cdot)$ is a probability distribution; (ii) from x_0 with $h_0(x_0) > 0$,*

$$\mathbb{P}^g(X_K = y \mid x_0) = \mathbb{P}(X_K = y \mid x_0) \frac{g(y)}{h_0(x_0)}, \quad h_0(x_0) = \sum_y \mathbb{P}(X_K = y \mid x_0) g(y), \quad (5)$$

the exact base terminal law tilted by g , normalized by the full-budget value at the start; (iii) $P_k^g(x, y) > 0 \Rightarrow P_k(x, y) > 0$ (controlled support $\subseteq$ base support); and (iv) states with $h_k(x) = 0$ carry zero controlled forward probability, so the ratio is never evaluated where undefined.

Proof sketch. (i) $\sum_y P_k(x, y) h_{k+1}(y) = h_k(x)$, so the ratios sum to one. (ii) Telescoping the controlled path measure cancels each interior h , leaving $\mathbb{P}(\text{path})g(y)/h_0(x_0)$; sum over interior paths. (iii)–(iv) the multiplier is finite and nonnegative where defined and can only rescale, never create, an edge; a zero-value state is entered with zero probability. Full time-inhomogeneous, zero- h , and event-conditioning details are Appendix A.5, checked numerically in Appendix E. $\square$

The transform is machine-exact on an enumerable molecular chain: the controlled terminal law matches the analytic tilt to total variation 8×10^{-17} , with no support violations.

Immediate consequences (Theorem 3). **Validity:** controlled support $\subseteq$ base support $\subseteq$ legal edits (Proposition 1), so every controlled state is valid—validity is inherited from the support, not created by the transform, and a controller cannot guide off the valid state space. **Exact events:** for $g = \mathbf{1}[x \in E]$ with E base-reachable, (5) is the base law conditioned on $\{X_K \in E\}$, realized exactly; an unreachable E leaves it undefined, as no controller can manufacture missing support.

Corollary 4 (Dynamic retargeting). *Replacing g by g' at x_τ and transforming the remaining budget $K - \tau$ yields the exact g' -tilt over that budget conditioned on x_τ —in general not the law from using g' at step 0, so continuation-versus-restart is a substantive question (verified exact, $TV 9 \times 10^{-17}$). Exactness is budget-specific, so the controller conditions on b .*

Corollary 5 (Pathwise constraints). *A constraint required at every committed state is path-dependent. When rule-closed (a state predicate partitioning marks into allowed/forbidden), it is enforced exactly by deleting violating marks from $\mathcal{A}(\cdot)$ —a support restriction giving a valid embedded chain to which Theorem 3 applies—so pathwise feasibility holds by construction; a genuinely cumulative constraint is handled by augmenting the state with the running statistic and conditioning on the augmented terminal event (Appendix A.7).*

6 THE GOAL-CONDITIONED CONTROLLER

Realizing Theorem 3 requires the value function h , in three regimes we evaluate separately. *Exact dynamic programming* applies where the reachable set from x_0 within budget K can be enumerated: we compute h by the backward recursion and steer with the exact controlled kernel. This is the regime in which the terminal tilt, event conditioning, and dynamic retargeting are all machine-exact (Section 5 and Appendix E); it is the ground truth for every approximate controller and is realized in the implementation. *Amortized learned control* applies at scale, where enumeration is impossible: a network $h_\phi(b, x; x_{\text{src}}, z, m) \approx h_{K-b}(x)$ conditions on the remaining budget b (essential; the exact value depends on it, Corollary 4), the source x_{src} (via source-relative features and the protected mask

m), and the goal z . Its regression target is the finite-horizon value of Theorem 3 itself—the expected terminal desirability under the base chain, $h_{K-b}(x) = \mathbb{E}[g_z(X_K) \mid X_{b\text{-to-go}} = x]$ —so we introduce no separate control objective; the estimator and parameterization are specified in Appendix C.1, where h_ϕ is evaluated as an approximation (value calibration, backward-equation residual, control quality against the exact controller), not assumed exact. Because it is approximate, h_ϕ is used self-normalized over the successor set,

$$P_b^\phi(x, y) = \frac{P_\theta(y \mid x) h_\phi(b-1, y; x_{\text{src}}, z, m)}{\sum_{y'} P_\theta(y' \mid x) h_\phi(b-1, y'; x_{\text{src}}, z, m)}, \quad (6)$$

a proper kernel on the base support for any nonnegative h_ϕ that reduces to Theorem 3 when $h_\phi = h$; it reweights P_θ and never adds support, so validity holds for the learned controller regardless of value error—an imperfect controller can misallocate probability but cannot produce an invalid molecule. When an unbiased terminal law is required despite value error, an *optional* twisted-SMC corrector runs a population under (6) with h_ϕ as the twist, asymptotically exact in the particle count (Wu et al., 2023; Skreta et al., 2025); we report its effective sample size rather than defaulting to it. In all cases the controller changes which legal edits fire, never the chemistry prior, so a preference switch is a controller swap (Corollary 4) and the base process is held fixed across every controller comparison.

7 MULTI-OBJECTIVE AND DYNAMIC CONTROL

Goal-conditioned, not scalarized. Linear scalarization $U_w(x) = w^\top F(x)$ undersamples non-convex fronts and slides to extremes, so instead of a fixed weight the descriptor z specifies an objective-space region $\mathcal{B}_z$ (a box, Chebyshev direction, or band) and the controller steers toward g_z (1). For a region indicator, $h_{K-b}(x) = \mathbb{P}(F(X_K) \in \mathcal{B}_z \mid X_{b\text{-to-go}} = x)$ (exact-events consequence of Theorem 3) is exactly the probability of reaching the region within the remaining budget, so a goal-conditioned value function is a reachability-to-region map and different z reuse the same base process.

Pareto exploration. We grow hypervolume directly rather than search over weights: maintaining the nondominated archive $\mathcal{A}$, we repeatedly (i) identify an under-covered or high-contribution region of the current front, (ii) set z to it, (iii) steer or branch a controlled trajectory toward it under a fixed base process, and (iv) fold the evaluated committed states into $\mathcal{A}$. Branching from a saved archive intermediate is exact continuation (Corollary 4): a shared prefix fans into several preference-conditioned trajectories without restarting (the “Pareto fan” of Section 9). A mid-run preference change is served by continuation, and a mid-run constraint injection by pathwise support restriction (Corollary 5).

One controlled steer (Algorithm 2) enumerates $\mathcal{A}(x)$, aggregates to $P_\theta(\cdot \mid x)$, restricts the support to marks respecting m and any pathwise constraint, reweights by (6), samples a successor, and folds it into $\mathcal{A}$; every scored state is a committed molecule. Universal edit-prior training, the exact enumerable controller, the single-steer routine, and the region-selecting exploration loop are Algorithms 1–3 in Appendix C, budgeted per Section 10.

8 RELATED WORK

Molecular generators and editing. Learned generators span latent graph models (Jin et al., 2018; 2020), RL atom/bond editors (Zhou et al., 2019), retrieval control (Wang et al., 2023), discrete-diffusion generalists serving hit and lead optimization (Lee et al., 2025; Gat et al., 2024; Kaeche et al., 2025), and graph-to-graph or one-fragment optimizers from paired supervision (Jin et al., 2019; Chen et al., 2021); matched molecular pairs supply the local transformations we use as supervision (Hussain & Rea, 2010; Dalke et al., 2018). We do not claim that editing valid molecules is new—fragment MCMC (Xie et al., 2021), graph genetic algorithms (Jensen, 2019), reaction GFlowNets (Kozłowski et al., 2024), structure-preserving diffusion editors (Feng et al., 2025), and reachability/world-model editors (Qiao et al., 2026) all move between complete molecules. Our distinction is a *learned generator-matched process* on an operator-closed molecular state space whose steering is a Doob transform, rather than heuristic proposals scored at the endpoint or a code that is a molecule only after decoding.

Multi-objective and Pareto methods. We adopt the metric and search machinery of evolutionary optimizers (Deb et al., 2002; Deb & Jain, 2014; Zhang & Li, 2007; Beume et al., 2007; Daulton et al., 2020) and its indicators (normalized hypervolume, IGD+, additive ϵ) (Zitzler & Thiele, 1999; Ishibuchi et al., 2015; Zitzler et al., 2003); preference/goal-conditioned GFlowNets and flow-matching optimizers sample Pareto-diverse candidates (Jain et al., 2023; Bengio et al., 2021; 2023; Yuan et al.,

2025; Chen et al., 2025a). For fairness we let such operators act on *our* legal proposal set (Section 9). MOG-DFM steers a pretrained discrete flow-matching generator by a rank-directional score and hypercone filter on peptide/DNA design (Chen et al., 2025b); it addresses *how to choose* among transitions, whereas we *define* the chemistry-native transition system and supply exact path control. The two are orthogonal: we place a MOG-DFM-style controller over our identical kernel in the same-base table, not as an external baseline.

Stochastic control and generative flows. Generator matching learns a marginal Markov generator from tractable conditional generators, subsuming diffusion, flow matching, and jump processes (Holderrieth et al., 2025; Lipman et al., 2023; Campbell et al., 2024; 2022; Lou et al., 2024); we instantiate it on an executable rewrite process, with rewriting semantics from graph-rewriting theory (Danos et al., 2015; Behr et al., 2021). Our controller is a finite-horizon Doob h -transform (Doob, 1984; Rogers & Williams, 2000), familiar from diffusion and Schrödinger bridges (Zhou et al., 2024; De Bortoli et al., 2021). It contrasts with reward/classifier guidance, which lacks a discrete distributional guarantee (Dhariwal & Nichol, 2021; Ho & Salimans, 2022); with reward and adjoint fine-tuning, which change *weights* rather than steer a fixed prior (Black et al., 2024; Fan et al., 2023; Domingo-Enrich et al., 2025; Wang et al., 2025; Zekri & Boullé, 2025); and with twisted-SMC/Feynman-Kac correctors, an optional asymptotically-exact knob (Wu et al., 2023; Skreta et al., 2025; Singhal et al., 2025; Del Moral, 2004; Doucet et al., 2001; Du & Mordatch, 2019). Conservatively, our contribution is that a learned generator-matched process over an operator-closed molecular state space makes exact, retargetable, support-preserving control well-defined and applicable throughout budgeted lead optimization; standardized corpora and metrics are used where noted (Brown et al., 2019; Gao et al., 2022; Irwin et al., 2012; Bickerton et al., 2012; Ertl & Schuffenhauer, 2009; Rogers & Hahn, 2010; Preuer et al., 2018; Guo & Schwaller, 2024b; Kim et al., 2024; Loeffler et al., 2024; Guo & Schwaller, 2024a).

9 EXPERIMENTS

We answer five questions. **RQ1 (exactness):** does the Doob transform realize a computable target law exactly, and where do reward-local controllers fail? **RQ2 (primary):** holding the process fixed, does control improve the multi-objective lead-optimization front at a matched budget, attributable to the control law? **RQ3 (dynamic):** can preferences change mid-trajectory without restarting? **RQ4 (path-wise):** can constraints hold at *every* committed molecule, more efficiently than endpoint filtering? **RQ5 (attribution/robustness):** which component causes the effect, and does it survive a held-out evaluator? Sources, objectives, budgets, and statistics are fixed in Section 10 before any result; every empirical cell comes from a result registry, shown here as a visible placeholder until its run completes. Implementation correctness (closure, kernel normalization, machine-exact Doob residual, quotient-kernel equality, tiny-corpus convergence, all-intermediate validity) is *implementation-validation* evidence (Appendix E), not a headline result.

RQ1 — Exact conditioning on an enumerable chain. *Protocol:* enumerate a bounded molecular embedded chain by breadth-first expansion of the real legal fiber from methane (4 heavy atoms, 572 states), compute the exact terminal law and value by the backward recursion of Theorem 3, and condition on a rare substructure, a property interval, two simultaneous constraints, a Boltzmann tilt, and a region, plus mid-trajectory retargeting. *Baselines:* exact Doob, learned controller, greedy ΔF , local Boltzmann, classifier-value guidance, endpoint reranking, MOG-DFM-style, SMC corrector—all on the identical kernel. *Metrics:* total variation to the exact law, target-event error, value calibration and backward residual, support violations. *Result:* On the enumerable chain the exact Doob transform realizes the analytic target law to within `xxx` total variation and its terminal-tilt derivation is machine-exact (8×10^{-17}), whereas one-step reward-greedy control attains `xxx` and endpoint reranking `xxx`; the sign and magnitude of this gap determine whether reward-local steering recovers the correct conditional distribution or only its high-reward mode. (panels in Appendix F). *Interpretation:* a near-zero exact-Doob TV against a larger greedy TV means reward-local control finds high reward at the wrong distribution. *Limitation:* exactness is shown only where enumeration is possible.

RQ2 — Multi-objective similarity-constrained lead optimization (primary). *Protocol:* from a provenance-documented lead set (The lead set comprises `xxx` sources drawn from `xxx`; its provenance and license are documented in the data appendix, and it is used with a primary citation rather than as an unattributed internal benchmark.), each method optimizes an objective vector (predicted affinity, drug-likeness, a synthetic-tractability surrogate, source similarity) under a fixed budget, reference point, archive cap, and identical constraints; we report a vector-valued front, not a best-docking scalar. *Baselines:* the same-base controller hierarchy (unguided, endpoint+rerank, greedy, local

Table 1: Multi-objective similarity-constrained lead optimization (RQ2): same-base controller comparison (aggregate over sources at a matched oracle budget). All rows share the base process, so differences isolate the control law. Arrows give the improving direction; each cell reports the point estimate, with paired bootstrap 95% intervals (resampling unit: source) in Appendix F. Reference point, objective bounds, and archive cap are fixed before evaluation (Section 10); no cell is emphasized. External optimizers (GraphGA, MARS, GenMol, RetMol) are contextual and reported in Table 3. Placeholders (xxx) are filled from the registry when the runs complete.

Controller (same base)	nHV $\uparrow$	HV-AUC $\uparrow$	IGD+ $\downarrow$	#Feas $\uparrow$	Src-sim $\uparrow$	Orc-eff $\uparrow$
Unguided edit prior	xxx	xxx	xxx	xxx	xxx	xxx
Endpoint + rerank	xxx	xxx	xxx	xxx	xxx	xxx
Greedy ΔF	xxx	xxx	xxx	xxx	xxx	xxx
Local Boltzmann	xxx	xxx	xxx	xxx	xxx	xxx
MOG-DFM-style	xxx	xxx	xxx	xxx	xxx	xxx
SMC / Feynman-Kac	xxx	xxx	xxx	xxx	xxx	xxx
Learned Doob (ours)	xxx	xxx	xxx	xxx	xxx	xxx

Boltzmann, MOG-DFM-style, SMC, learned Doob) plus external optimizers (GraphGA, MARS, GenMol, RetMol), documented per baseline in Appendix F. *Metrics*: normalized hypervolume and HV-AUC vs oracle calls (primary); IGD+, #feasible (secondary); source similarity, diversity, oracle efficiency, with paired bootstrap CIs over sources. *Result*: Table 1; Holding the molecular process fixed and varying only the controller, the learned Doob controller attains normalized hypervolume xxx against xxx for unguided editing and xxx for greedy control at a matched oracle budget, while retaining source similarity xxx; the ordering of these values against the same-base baselines determines whether the improvement is attributable to the control law rather than the state space. *Interpretation*: the load-bearing comparison is same-base—exceeding the other controllers on the identical kernel isolates the control law; external optimizers on our proposal set separate law from vocabulary. *Limitation*: better computational hypervolume is not experimental efficacy.

RQ3 — Dynamic preference steering. *Protocol*: preferences switch (potency+drug-likeness $\rightarrow$ solubility+safety $\rightarrow$ structural preservation) at fixed fractions; adaptation regret is defined mathematically in Appendix F before it is reported. *Baselines*: static compromise, continue-unchanged, restart-from-lead, restart-with-a-conventional-optimizer, greedy, MOG-DFM-style, endpoint rerank. *Metrics*: steps-to-region, adaptation regret, retention of prior gains, final hypervolume, oracle cost vs restart. *Result*: After a mid-trajectory preference switch, dynamic continuation reaches the new target region in xxx edits with adaptation regret xxx, against xxx for restarting from the lead, at xxx the oracle cost; whether continuation dominates restart is read directly from these paired quantities. The Pareto fan—one lead, a shared prefix, several preference-conditioned valid branches, every molecule from a real run—is Figure 4 (rendered in Appendix F and promoted to the main text once the run completes). *Interpretation*: continuation at lower regret and cost than restart shows budget-conditioned control adapts without discarding progress. *Limitation*: regret is measured against the switch-time archive, not long-run optimality.

RQ4 — Pathwise (not endpoint-only) constraints. *Protocol*: state-defined constraints (never alter a protected scaffold; never realize a forbidden substructure; stay in a size/charge range; keep $\text{sim}(x, x_{\text{src}}) \geq \tau$) hold throughout generation. *Baselines*: hard operator-support masking, Feynman-Kac running penalty, terminal penalty, endpoint rejection. *Metrics*: completely-feasible-trajectory fraction, integrated and maximum violation, final quality, feasible hypervolume, oracle calls wasted on rejected trajectories. *Result*: For a rule-closed pathwise constraint, legal-support restriction yields xxx completely-feasible trajectories by construction, against xxx for endpoint filtering, which wastes xxx of its oracle budget on trajectories rejected only at the end. *Interpretation*: a rule-closed constraint is exact under support restriction by construction (Corollary 5); the question is how much budget endpoint filtering wastes. *Limitation*: only rule-closed constraints are exact; cumulative ones need state augmentation.

RQ5 — Causal ablations and oracle robustness. *Protocol*: decompose the system along its four factors (valid operator support, learned prior, control law, goal policy), and separately optimize one oracle while evaluating with an independent one at matched out-of-distribution distance. *Baselines*: full system; uniform legal proposals for the prior; control removed; goal policy replaced by fixed scalarization; a known-buggy legacy sampler is *not* revived (appendix diagnostic only, Appendix E).

Metrics: normalized hypervolume per ablation; guidance-to-evaluator gap, uncertainty, OOD distance, alert statistics. *Result:* Decomposing the system, normalized hypervolume moves from xxx (edit prior, no control) to xxx (full system), while removing the learned prior in favor of uniform legal proposals gives xxx; the relative sizes of these steps attribute the effect to the operator support, the learned prior, and the control law respectively. Under an independent held-out evaluator the guidance-to-evaluator score gap is xxx and the re-scored hypervolume is xxx; the size of the degradation relative to the in-oracle result bounds how much of the apparent gain survives a change of evaluator. *Interpretation:* the ablation attributes the effect to components; the held-out gap bounds oracle exploitation, to which a flexible legal editor is exactly the vulnerable optimizer. *Limitation:* held-out evaluators are themselves imperfect. As a secondary demonstration that the operators form a genuine generative process, de novo carbon-tree generation (validity, uniqueness, novelty, FCD; keys DENOVO_VALIDITY, DENOVO_FCD) is reported in Appendix F, on which the paper’s claims do not depend.

10 FAIRNESS, BUDGETS, AND STATISTICS

We fix the protocol before any result. *Matched conditions:* within a question every method receives identical sources, objective vector and bounds, hard constraints and protected masks, oracle-call budget, per-method archive cap, and a single published hypervolume reference point, with normalization fixed in advance; the load-bearing comparison holds the process fixed and varies only the controller (Table 1), while external optimizers are contextual and, where possible, additionally run on our legal proposal set. *Oracle accounting:* we count every oracle evaluation—on committed intermediates, rejected proposals, invalid or repaired candidates, reranked candidates, and all SMC particles—and report hypervolume against oracle calls, not only its final value; we separately report, never conflating, oracle calls, forward passes, successors scored, accepted edits, and wall-clock/GPU-hours, so a controller that scores many successors per accepted edit is charged for them. *Statistics:* each configuration runs over multiple seeds, comparisons are paired across identical sources, and we report paired bootstrap 95% intervals with the resampling unit stated per metric (source, target-event, or trajectory), both per-task and aggregate. *Selection hygiene:* no checkpoint, hyperparameter, or preference schedule is chosen on the test set, and any asymmetric tuning is disclosed. These rules are why the registry stores a resampling unit and interval for every cell.

11 LIMITATIONS

Graph validity is not synthetic feasibility: our operators guarantee connected, valence-admissible molecules, not synthesizable ones (we say “graph-valid”, never “synthesizable”), and the representation excludes stereochemistry, isotopes, radicals, and charge-changing edits—we preserve a source charge state rather than design over charge. The operator vocabulary bounds the reachable chemistry, and support-preserving control cannot manufacture a transformation no enabled edit realizes. Exactness holds only where the value function is exact (enumerable spaces); at scale the amortized controller is an approximation we measure rather than assume, the oracles are exploitable surrogates evaluated against an independent one, and a better computational front is not experimental efficacy. External optimizers differ in scale and representation, so the controller comparison is the attributable one (extended limitations in Appendix F).

12 CONCLUSION

We cast source-conditioned molecular optimization as stochastic control of a generator-matched jump process on an operator-closed space of connected, valence-valid molecular graphs, and derived a finite-horizon Doob transform that steers the fixed-budget editing chain while preserving its legal support, realizing the exact terminal tilt under an exact value function, and admitting exact mid-trajectory retargeting and pathwise constraints. Because every state is a molecule and every preference update changes the controller rather than the prior, one held-fixed process supports similarity-constrained multi-objective lead optimization, dynamic steering, and hard pathwise constraints—establishing executable rewriting as a control substrate for generator matching, and pathwise chemical closure as a substrate for exact, oracle-efficient molecular design.

ETHICS STATEMENT

This work concerns computational molecular optimization. Its intended positive impact is more sample-efficient, constraint-respecting lead optimization that reduces wasted oracle and synthesis effort. As with any molecular generator, the same machinery could in principle be directed toward harmful compounds; our method optimizes user-specified objectives and does not itself target toxicity, and its outputs are graph-valid but not established to be synthesizable, active, or safe—every candidate would require standard experimental and regulatory evaluation before any use. We use public benchmark corpora and computational property surrogates only, with no human subjects, no personal data, and no wet-lab experiments. The pathwise-constraint machinery can be used to *exclude* forbidden substructures throughout generation, which we view as a safety-relevant capability. We disclose the use of learned property oracles and their exploitability (Section 11) so that reported computational gains are not mistaken for demonstrated biological efficacy. A separate, factual account of large language model use in preparing this work is given in Appendix G.

REPRODUCIBILITY STATEMENT

This statement maps claims to the evidence needed to reproduce them. The state space, operator registry, and closure argument are Section 3 and Appendices A.2 and B. The generator-matching objective and the canonical-kernel normalization, with their measure identities, are Appendices A.3 and A.4. The finite-horizon Doob transform and its corollaries are proved in Appendices A.5 to A.7, and every finite-state identity is checked numerically against the implemented process in Appendix E, which also records the enumerable-benchmark construction and tolerances. The training corpus, its broad-organic scope and census, the standardization and rejection rules, the split protocol, the matched-molecular-pair and scaffold mining, the corruption and trace compiler, the anti-concentration safeguards, the data-driven curriculum, and all scaled counts are Appendix D; the base checkpoint, head expansion, warm-start, optimizer, schedule, and checkpoint selection are documented there as well. The controller design, its regression target, and its evaluation-as-approximation are Appendix C.1. The fairness, budgeting, and statistics protocol is Section 10. Every empirical value is drawn from a result registry keyed to a machine-readable artifact; a released version will include the registry, the plotting artifacts, and the evaluation harness. An anonymous-release checklist—scrubbing author and user names, home-directory paths, cloud profile names, repository remotes, third-person self-citations, and PDF metadata—is Appendix H.

REFERENCES

- Nicolas Behr, Vincent Danos, and Ilias Garnier. Stochastic mechanics of graph rewriting. *arXiv preprint arXiv:2003.02132*, 2021.
- Emmanuel Bengio, Moksh Jain, Maksym Korablyov, Doina Precup, and Yoshua Bengio. Flow network based generative models for non-iterative diverse candidate generation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021. arXiv:2106.04399.
- Yoshua Bengio, Salem Lahlou, Tristan Deleu, Edward J. Hu, Mo Tiwari, and Emmanuel Bengio. GFlowNet foundations. *Journal of Machine Learning Research*, 24, 2023.
- Nicola Beume, Boris Naujoks, and Michael Emmerich. SMS-EMOA: Multiobjective selection based on dominated hypervolume. *European Journal of Operational Research*, 181(3):1653–1669, 2007.
- G. Richard Bickerton, Gaia V. Paolini, Jérémy Besnard, Sorel Muresan, and Andrew L. Hopkins. Quantifying the chemical beauty of drugs. *Nature Chemistry*, 4(2):90–98, 2012.
- Kevin Black, Michael Janner, Yilun Du, Ilya Kostrikov, and Sergey Levine. Training diffusion models with reinforcement learning. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2305.13301.
- Nathan Brown, Marco Fiscato, Marwin H. S. Segler, and Alain C. Vaucher. Guacamol: Benchmarking models for de novo molecular design. *Journal of Chemical Information and Modeling*, 59(3): 1096–1108, 2019.

- Andrew Campbell, Joe Benton, Valentin De Bortoli, Tom Rainforth, George Deligiannidis, and Arnaud Doucet. A continuous time framework for discrete denoising models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. arXiv:2205.14987.
- Andrew Campbell, Jason Yim, Regina Barzilay, Tom Rainforth, and Tommi Jaakkola. Generative flows on discrete state-spaces: Enabling multimodal flows with applications to protein co-design. In *International Conference on Machine Learning (ICML)*, 2024. arXiv:2402.04997.
- Tong Chen, Yinuo Zhang, and Pranam Chatterjee. AReURDi: Annealed rectified updates for refining discrete flows with multi-objective guidance. *arXiv preprint arXiv:2510.00352*, 2025a.
- Tong Chen, Yinuo Zhang, Sophia Tang, and Pranam Chatterjee. Multi-objective-guided discrete flow matching for controllable biological sequence design. *arXiv preprint arXiv:2505.07086*, 2025b. ICML 2025 GenBio Workshop (spotlight).
- Ziqi Chen, Martin Renqiang Min, Srinivasan Parthasarathy, and Xia Ning. A deep generative model for molecule optimization via one fragment modification. *Nature Machine Intelligence*, 3(12): 1040–1049, 2021.
- Andrew Dalke, Jérôme Hert, and Christian Kramer. mmpdb: An open-source matched molecular pair platform for large multiproperty data sets. *Journal of Chemical Information and Modeling*, 58(5):902–910, 2018.
- Vincent Danos, Tobias Heindel, Ricardo Honorato-Zimmer, and Sandro Stucki. Thermodynamic graph rewriting. *Logical Methods in Computer Science*, 2015.
- Samuel Daulton, Maximilian Balandat, and Eytan Bakshy. Differentiable expected hypervolume improvement for parallel multi-objective bayesian optimization. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. arXiv:2006.05078.
- Valentin De Bortoli, James Thornton, Jeremy Heng, and Arnaud Doucet. Diffusion schrödinger bridge with applications to score-based generative modeling. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021. arXiv:2106.01357.
- Kalyanmoy Deb and Himanshu Jain. An evolutionary many-objective optimization algorithm using reference-point-based nondominated sorting approach, part i: Solving problems with box constraints. *IEEE Transactions on Evolutionary Computation*, 18(4):577–601, 2014.
- Kalyanmoy Deb, Amrit Pratap, Sameer Agarwal, and T. Meyarivan. A fast and elitist multiobjective genetic algorithm: NSGA-II. *IEEE Transactions on Evolutionary Computation*, 6(2):182–197, 2002.
- Pierre Del Moral. *Feynman-Kac Formulae: Genealogical and Interacting Particle Systems with Applications*. Springer, 2004.
- Prafulla Dhariwal and Alexander Nichol. Diffusion models beat GANs on image synthesis. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021. arXiv:2105.05233.
- Carles Domingo-Enrich, Michal Drozdal, Brian Karrer, and Ricky T. Q. Chen. Adjoint matching: Fine-tuning flow and diffusion generative models with memoryless stochastic optimal control. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2409.08861.
- Joseph L. Doob. *Classical Potential Theory and Its Probabilistic Counterpart*. Springer-Verlag, 1984.
- Arnaud Doucet, Nando de Freitas, and Neil Gordon (eds.). *Sequential Monte Carlo Methods in Practice*. Springer, 2001.
- Yilun Du and Igor Mordatch. Implicit generation and modeling with energy-based models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019. arXiv:1903.08689.
- Peter Ertl and Ansgar Schuffenhauer. Estimation of synthetic accessibility score of drug-like molecules based on molecular complexity and fragment contributions. *Journal of Cheminformatics*, 1:8, 2009.

- Ying Fan, Olivia Watkins, Yuqing Du, Hao Liu, Moonkyung Ryu, Craig Boutilier, Pieter Abbeel, Mohammad Ghavamzadeh, Kangwook Lee, and Kimin Lee. DPOK: Reinforcement learning for fine-tuning text-to-image diffusion models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. arXiv:2305.16381.
- Yuanxin Feng et al. MolEditRL: Structure-preserving molecular editing via discrete diffusion and reinforcement learning. *arXiv preprint arXiv:2505.20131*, 2025.
- Wenhao Gao, Tianfan Fu, Jimeng Sun, and Connor W. Coley. Sample efficiency matters: A benchmark for practical molecular optimization. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2022. arXiv:2206.12411.
- Itai Gat, Tal Remez, Neta Shaul, Felix Kreuk, Ricky T. Q. Chen, Gabriel Synnaeve, Yossi Adi, and Yaron Lipman. Discrete flow matching. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. arXiv:2407.15595.
- Jeff Guo and Philippe Schwaller. Augmented memory: Sample-efficient generative molecular design with reinforcement learning. *JACS Au*, 4(6):2160–2172, 2024a.
- Jeff Guo and Philippe Schwaller. Saturn: Sample-efficient generative molecular design using memory manipulation. *arXiv preprint arXiv:2405.17066*, 2024b.
- Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance. *arXiv preprint arXiv:2207.12598*, 2022.
- Peter Holderrieth, Marton Havasi, Jason Yim, Neta Shaul, Itai Gat, Tommi Jaakkola, Brian Karrer, Ricky T. Q. Chen, and Yaron Lipman. Generator matching: Generative modeling with arbitrary markov processes. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2410.20587.
- Jameed Hussain and Ceara Rea. Computationally efficient algorithm to identify matched molecular pairs (mmps) in large data sets. *Journal of Chemical Information and Modeling*, 50(3):339–348, 2010.
- John J. Irwin, Teague Sterling, Michael M. Mysinger, Erin S. Bolstad, and Ryan G. Coleman. ZINC: A free tool to discover chemistry for biology. *Journal of Chemical Information and Modeling*, 52(7):1757–1768, 2012.
- Hisao Ishibuchi, Hiroyuki Masuda, Yuki Tanigaki, and Yusuke Nojima. Modified distance calculation in generational distance and inverted generational distance. In *Evolutionary Multi-Criterion Optimization (EMO)*, 2015.
- Moksh Jain, Sharath Chandra Raparthy, Alex Hernández-García, Jarrid Rector-Brooks, Yoshua Bengio, Santiago Miret, and Emmanuel Bengio. Multi-objective gflownets. In *International Conference on Machine Learning (ICML)*, 2023. arXiv:2210.12765.
- Jan H. Jensen. A graph-based genetic algorithm and generative model/monte carlo tree search for the exploration of chemical space. *Chemical Science*, 10(12):3567–3572, 2019.
- Wengong Jin, Regina Barzilay, and Tommi Jaakkola. Junction tree variational autoencoder for molecular graph generation. In *International Conference on Machine Learning (ICML)*, 2018. arXiv:1802.04364.
- Wengong Jin, Kevin Yang, Regina Barzilay, and Tommi Jaakkola. Learning multimodal graph-to-graph translation for molecular optimization. In *International Conference on Learning Representations (ICLR)*, 2019. arXiv:1812.01070.
- Wengong Jin, Regina Barzilay, and Tommi Jaakkola. Hierarchical generation of molecular graphs using structural motifs. In *International Conference on Machine Learning (ICML)*, 2020. arXiv:2002.03230.
- Benno Kaeck, Luis Wyss, Karsten Borgwardt, and Gianvito Grasso. Refine drugs, don’t complete them: Uniform-source discrete flows for fragment-based drug discovery. *arXiv preprint arXiv:2509.26405*, 2025.

- Hyeonah Kim, Minsu Kim, Sanghyeok Choi, and Jinkyoo Park. Genetic-guided gflownets for sample efficient molecular optimization. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. arXiv:2402.05961.
- Michał Koziarski, Andrei Rekes, Dmytro Shevchuk, Almer van der Sloot, Piotr Gaiński, Yoshua Bengio, Cheng-Hao Liu, Mike Tyers, and Robert Batey. RGFN: Synthesizable molecular generation using gflownets. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. arXiv:2406.08506.
- Seul Lee, Karsten Kreis, Srimukh Prasad Veccham, Meng Liu, Danny Reidenbach, Yuxing Peng, Saeed Paliwal, Weili Nie, and Arash Vahdat. GenMol: A drug discovery generalist with discrete diffusion. In *International Conference on Machine Learning (ICML)*, 2025. arXiv:2501.06158.
- Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling. In *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2210.02747.
- Hannes H. Loeffler, Jiazhen He, Alessandro Tibo, Jon Paul Janet, Alexey Voronov, Lewis H. Mervin, and Ola Engkvist. Reinvent 4: Modern ai-driven generative molecule design. *Journal of Cheminformatics*, 16:20, 2024.
- Aaron Lou, Chenlin Meng, and Stefano Ermon. Discrete diffusion modeling by estimating the ratios of the data distribution. In *International Conference on Machine Learning (ICML)*, 2024. arXiv:2310.16834.
- Kristina Preuer, Philipp Renz, Thomas Unterthiner, Sepp Hochreiter, and Günter Klambauer. Fréchet chemnet distance: A metric for generative models for molecules in drug discovery. *Journal of Chemical Information and Modeling*, 58(9):1736–1741, 2018.
- Yang Qiao, Bo Pan, Hao-Wei Pang, Peter Zhiping Zhang, Liying Zhang, and Liang Zhao. MolWorld: Molecule world models for actionable molecular optimization. *arXiv preprint arXiv:2605.08954*, 2026.
- David Rogers and Mathew Hahn. Extended-connectivity fingerprints. *Journal of Chemical Information and Modeling*, 50(5):742–754, 2010.
- L. C. G. Rogers and David Williams. *Diffusions, Markov Processes and Martingales, Volume 1: Foundations*. Cambridge University Press, 2nd edition, 2000.
- Raghav Singhal, Zachary Horvitz, Ryan Teehan, Mengye Ren, Zhou Yu, Kathleen McKeown, and Rajesh Ranganath. A general framework for inference-time scaling and steering of diffusion models. *arXiv preprint arXiv:2501.06848*, 2025.
- Marta Skreta, Tara Akhound-Sadegh, Viktor Ohanesian, Roberto Bondesan, Alán Aspuru-Guzik, Arnaud Doucet, Rob Brekelmans, Alexander Tong, and Kirill Neklyudov. Feynman-kac correctors in diffusion: Annealing, guidance, and product of experts. In *International Conference on Machine Learning (ICML)*, 2025. arXiv:2503.02819.
- Chenyu Wang, Masatoshi Uehara, Yichun He, Amy Wang, Tommaso Biancalani, Avantika Lal, Tommi Jaakkola, Sergey Levine, Hanchen Wang, and Aviv Regev. Fine-tuning discrete diffusion models via reward optimization with applications to DNA and protein design. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2410.13643.
- Zichao Wang, Weili Nie, Zhuoran Qiao, Chaowei Xiao, Richard G. Baraniuk, and Anima Anandkumar. Retrieval-based controllable molecule generation. In *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2208.11126.
- Luhuan Wu, Brian L. Trippe, Christian A. Naesseth, David M. Blei, and John P. Cunningham. Practical and asymptotically exact conditional sampling in diffusion models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. arXiv:2306.17775.
- Yutong Xie, Chence Shi, Hao Zhou, Yuwei Yang, Weinan Zhang, Yong Yu, and Lei Li. MARS: Markov molecular sampling for multi-objective drug discovery. In *International Conference on Learning Representations (ICLR)*, 2021. arXiv:2103.10432.

- Ye Yuan, Can Chen, Christopher Pal, and Xue Liu. ParetoFlow: Guided flows in multi-objective optimization. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2412.03718.
- Oussama Zekri and Nicolas Boullé. Fine-tuning discrete diffusion models with policy gradient methods. *arXiv preprint arXiv:2502.01384*, 2025.
- Qingfu Zhang and Hui Li. MOEA/D: A multiobjective evolutionary algorithm based on decomposition. *IEEE Transactions on Evolutionary Computation*, 11(6):712–731, 2007.
- Linqi Zhou, Aaron Lou, Samar Khanna, and Stefano Ermon. Denoising diffusion bridge models. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2309.16948.
- Zhenpeng Zhou, Steven Kearnes, Li Li, Richard N. Zare, and Patrick Riley. Optimization of molecules via deep reinforcement learning. *Scientific Reports*, 9:10752, 2019.
- Eckart Zitzler and Lothar Thiele. Multiobjective evolutionary algorithms: A comparative case study and the strength pareto approach. *IEEE Transactions on Evolutionary Computation*, 3(4):257–271, 1999.
- Eckart Zitzler, Lothar Thiele, Marco Laumanns, Carlos M. Fonseca, and Viviane Grunert da Fonseca. Performance assessment of multiobjective optimizers: An analysis and review. *IEEE Transactions on Evolutionary Computation*, 7(2):117–132, 2003.

A PROOFS

Every finite-state identity below is additionally verified numerically against the implemented process on the enumerable benchmark (Appendix E); we state the residual tolerances there. We do not replace any central proof with an observed validity rate.

A.1 STATE SPACE

The state space $\mathcal{X}_{V,N}$ is the set of connected molecular graphs over the declared vocabulary with at most N heavy atoms in which every atom v satisfies the valence identity $\sum_u \text{ord}(v, u) + \text{H}(v) - \text{charge}(v) \in \text{Val}(\text{elem}(v))$, where bond orders are integers (Kekulé) and $\text{H}(v)$ is the implicit-hydrogen count. $\mathcal{X}_{V,N}$ is finite: elements, charges, hydrogen counts, and bond orders each range over finite sets, and the heavy-atom count is bounded by N , so the number of labeled graphs is finite and the quotient by graph isomorphism is finite. It is a discrete set with no metric or differential structure assumed; we never treat it as a manifold. Two graphs are identified iff they are isomorphic as labeled molecular graphs after canonicalization; membership, equality, and the executor are all defined on these canonical representatives.

A.2 OPERATOR CLOSURE (PER FAMILY)

We prove Proposition 1: for $x \in \mathcal{X}_{V,N}$ and $a \in \mathcal{A}(x)$, $T(x, a) \in \mathcal{X}_{V,N}$. Recall $\mathcal{A}(x)$ is the set of marks admitted by the executor, which (i) validates that $x \in \mathcal{X}_{V,N}$, (ii) checks the family’s application condition at the concrete match, (iii) executes the local rewrite, and (iv) re-validates the successor for per-atom valence, connectivity, and RDKit sanitizability, admitting the mark only if all pass and the successor is not isomorphic to x . Since admission requires (iv), it suffices to show each family’s rewrite *can* produce a valid successor and that the guard rejects the rest; we show the former family by family, so that $\mathcal{A}(x)$ is nonempty exactly where a valid edit exists.

atom_insert / atom_delete. A peripheral atom is added with an incident bond of declared order, or a peripheral atom (degree one, or a leaf whose removal keeps the graph connected) is removed; the touched neighbor’s implicit-hydrogen count is re-derived from its valence class so its valence identity holds, and all other atoms are unchanged. Heavy-atom count changes by ± 1 , staying within N by the guard. *atom_restate*. An atom’s valence class is changed among the classes feasible at its current heavy-bond sum; the implicit-hydrogen count is re-derived, preserving the valence identity, and no bond or other atom changes, so connectivity is preserved. *bond_reorder*. A non-ring bond’s order changes; the two endpoints’ hydrogen counts re-derive to preserve their valence identities; the edge support is unchanged, so connectivity is preserved. *bond_reroute (Graft)*. A pendant subtree attached across a single bridging bond is detached and reattached at a new site; the operation is lowered

to a delete of the bridging bond followed by an insert at the target, each validity-checked, and the moved fragment is internally unchanged, so valence identities hold and the result is connected by construction (the fragment reconnects to the retained core). *ring_system_grow / _delete / _restate*. A whole supported ring system is added, opened, or restated using a versioned typed ring catalog whose templates carry per-atom identity, ring topology, and aromatic/saturated role; each template instantiation is lowered to validity-checked primitive edits, the opened or added atoms re-derive hydrogen counts, and the successor is re-validated for connectivity. In every family untouched atoms are unchanged and touched atoms re-derive their hydrogen counts to satisfy the valence identity; the connectivity re-check in (iv) rejects any instantiation that would disconnect the graph. Hence every admitted mark yields $T(x, a) \in \mathcal{X}_{V,N}$, and closure is a property of the operators and the legality filter. $\square$

A.3 GENERATOR-MATCHING OBJECTIVE

We derive (3). The Poisson (generalized-KL) Bregman divergence between a model marked-rate measure $q_\theta(x \rightarrow a) = \Lambda_\theta(x)p_\theta(a | x)$ and a teacher measure q^* with a single nonzero mark a^* of rate r is $D(q^* || q_\theta) = \sum_a [q_\theta(a) - q^*(a) - q^*(a) \log(q_\theta(a)/q^*(a))] = \sum_a q_\theta(a) - r - r \log(q_\theta(a^*)/r)$, using $\sum_a q^*(a) = r$. Since p_θ normalizes, $\sum_a q_\theta(a) = \Lambda_\theta(x)$, and $\log q_\theta(a^*) = \log \Lambda_\theta(x) + \log p_\theta(a^* | x)$. Dropping the θ -independent constants $-r - r \log r$ leaves $\Lambda_\theta(x) - r[\log \Lambda_\theta(x) + \log p_\theta(a^* | x)]$, which is (3); at $r = 0$ only $\Lambda_\theta(x)$ remains. If $a^* \notin \mathcal{A}(x)$ then $p_\theta(a^* | x) = 0$ and $\log p_\theta = -\infty$, so the loss is non-finite—the guard that forces every teacher mark into the factorized fiber. The identity $\sum_a p_\theta(a | x) = 1$ (used above) is the normalization of the family-then-action softmax over the declared mark space, verified numerically to $\sim 10^{-7}$ (Appendix E). $\square$

A.4 CANONICAL QUOTIENT

We prove Proposition 2. The map $\rho : a \mapsto [T(x, a)]$ sends each legal mark to the isomorphism class of its executed successor. Isomorphism classes are disjoint, and every $a \in \mathcal{A}(x)$ has exactly one image (the executor is a function and self-loops—marks with $T(x, a) \cong x$ —are excluded from $\mathcal{A}(x)$), so $\{\rho^{-1}(y)\}_y = \{\mathcal{G}_y(x)\}_y$ partitions $\mathcal{A}(x)$. Therefore $\sum_y P_\theta(y | x) = \sum_y \sum_{a \in \mathcal{G}_y(x)} p_\theta(a | x) = \sum_{a \in \mathcal{A}(x)} p_\theta(a | x) = 1$. *Graft supervision*. The Graft family is scored at the successor group: its loss contribution uses $\log \sum_{a \in \mathcal{G}_y(x)} \exp(\cdot)$ (group-logsumexp), so generator matching supervises the molecular transition rate $P_\theta(y | x)$, not an individual coordinate. For non-Graft families, distinct coordinates mapping to one successor (e.g. deleting either of two symmetric methyls) each carry their own logit under a per-coordinate normalizer, so their probabilities *sum* to the successor rate—no over-count. This is why the successor sampler draws Graft one-representative-per-group but non-Graft per-coordinate; both induce the same kernel P_θ , verified equal to the raw sampler to $< 10^{-5}$ (Appendix E). $\square$

A.5 FINITE-HORIZON DOOB TRANSFORM (ALL EDGE CASES)

We prove Theorem 3 for a time-inhomogeneous sequence $P_0, \dots, P_{K-1}$, with h_k and P_k^g as defined in the theorem (backward recursion $h_k = P_k h_{k+1}$, $h_K = g$; controlled kernel $P_k^g(x, y) = P_k(x, y)h_{k+1}(y)/h_k(x)$ on $\{h_k > 0\}$). (i) *Stochasticity*. For x with $h_k(x) > 0$, $\sum_y P_k^g(x, y) = \frac{1}{h_k(x)} \sum_y P_k(x, y)h_{k+1}(y) = 1$, and each summand is ≥ 0 . (iv) *Zero-value boundary*. If $h_k(x) = 0$ then, since $P_k, h_{k+1} \geq 0$, $\sum_y P_k(x, y)h_{k+1}(y) = 0$ forces $P_k(x, y)h_{k+1}(y) = 0$ for all y . By induction from x_0 with $h_0(x_0) > 0$, the controlled one-step probability to y is $\propto P_k(x, y)h_{k+1}(y)$, zero whenever $h_{k+1}(y) = 0$; hence every reachable state has positive h and the controlled kernel is never evaluated where undefined. (ii) *Terminal law*. For a path $x_0, \dots, x_K = y$ with all $h_k(x_k) > 0$, the controlled path probability telescopes: $\prod_{k=0}^{K-1} P_k^g(x_k, x_{k+1}) = (\prod_k P_k(x_k, x_{k+1}))h_K(y)/h_0(x_0)$, since interior h cancel. As $h_K(y) = g(y)$, summing over interior paths gives (5), with $h_0(x_0) = \sum_y \mathbb{P}(X_K = y | x_0)g(y)$ by unrolling the recursion (Chapman–Kolmogorov); paths through a zero-value state contribute zero on both sides by (iv). (iii) *Support*. $P_k^g(x, y) > 0$ requires $P_k(x, y) > 0$ as the multiplier is finite. *Event conditioning*. For $g = \mathbf{1}[\cdot \in E]$, $h_0(x_0) = \mathbb{P}(X_K \in E | x_0)$ and the terminal law is $\mathbb{P}(X_K = y | x_0, X_K \in E)$; undefined iff E is unreachable. *CTMC-vs-fixed-step*. This transforms the embedded jump chain (the hazard is discarded in editing), not the continuous-time generator, and h is indexed by the discrete step / remaining budget, not a continuous time. $\square$

A.6 DYNAMIC RETARGETING

We prove Corollary 4. Fix an intermediate state x_τ , replace g by g' , and define h'_j by the backward recursion over the remaining $K - \tau$ steps with $h'_{K-\tau} = g'$. Applying Theorem 3 to the shifted chain $P_\tau, \dots, P_{K-1}$ started at x_τ gives the continuation terminal law $\mathbb{P}(X_K = y \mid x_\tau)g'(y)/h'_0(x_\tau)$: the exact g' -tilt over the remaining budget *conditioned on* x_τ . This differs from the tilt obtained by using g' from step 0, whose value functions propagate from x_0 ; the two agree only if x_τ has probability one under the from-0 tilt at step τ , which is not generic. Hence continuation and restart are distinct laws, and the controller must condition on the remaining budget. Budget-specificity: for $j < K - \tau$ the controlled marginal at step $\tau + j$ is $h'_{K-\tau-j}$ -weighted, *not* the terminal tilt unless $j = K - \tau$. Verified exact, TV 9×10^{-17} (Appendix E). $\square$

A.7 PATHWISE CONSTRAINTS AND STATE AUGMENTATION

We prove Corollary 5. *Rule-closed constraints.* Let $c : \mathcal{X}_{V,N} \rightarrow \{0, 1\}$ be a state predicate, requiring $c(X_k) = 1$ for all k . Define $\mathcal{A}(x)^c = \{a \in \mathcal{A}(x) : c(T(x, a)) = 1\}$ and the restricted kernel P^c from p_θ renormalized over $\mathcal{A}(x)^c$ (a valid embedded chain by the partition argument of Proposition 2 when $\mathcal{A}(x)^c \neq \emptyset$; an empty $\mathcal{A}(x)^c$ is terminal). Every trajectory of P^c satisfies c at every committed state by construction, and Theorem 3 applied to P^c gives the exact constraint-restricted terminal tilt; feasibility is a property of the support, so it is exact regardless of value error. *Cumulative constraints.* For a path functional $\phi_k = \Phi(\phi_{k-1}, X_k)$ (running maximum violation, accumulated cost, visited-motif set), augment $\tilde{X}_k = (X_k, \phi_k)$; the augmented process is Markov with kernel $\tilde{P}_k((x, \phi), (y, \phi')) = P_k(x, y)\mathbf{1}[\phi' = \Phi(\phi, y)]$, and a path-dependent event becomes a terminal event on $\tilde{X}_K$, to which Theorem 3 applies. The augmentation is exactly what turns a path functional into a terminal one; its cost is the enlarged state space. $\square$

A.8 APPROXIMATION ANALYSIS

On non-enumerable spaces the amortized h_ϕ replaces h in (6), which is self-normalized. Two facts bound the consequences. *Validity is unconditional:* (6) is supported on the base successor set for any nonnegative h_ϕ , so the support and validity consequences of Theorem 3 hold regardless of value error—an approximate controller misallocates probability but never leaves the valid state space. *Bias vanishes with an exact value:* (6) equals the exact controlled kernel of Theorem 3 when $h_\phi = h$, so the terminal-law error is controlled by the value error. We do not claim a closed-form bound on real spaces; instead we measure value calibration, the backward-equation residual, and total variation to the exact controller on the enumerable benchmark, and, when an unbiased law is required, restore asymptotic exactness with the twisted-SMC corrector (Section 6). We deliberately avoid asserting a distribution-level guarantee at scale, per Section 11.

B OPERATOR REGISTRY

Table 2 is the full registry of the 8 production operator families (of 10 defined; `cycle_insert`, `cycle_attach` are disabled everywhere and enumerate no candidates). Every family’s rewrite is lowered to validity-checked primitives; the “successor invariant” column states what the family guarantees beyond the universal connected-and-valence-valid closure of Proposition 1. Charge is preserved by every family (there is no charge-changing transition). “Training source” names which training layer typically supervises the family (Appendix D); ring-topology growth is retained from the de novo carbon-tree prior because the edit layers preserve ring count.

Disabled families. `cycle_insert`, `cycle_attach` are legacy null-prior operators retained in the mark enumeration for historical compatibility but enumerate no candidates and receive no training signal; ring additions go through `ring_system_grow`. *Charge.* No family alters an atom’s formal charge; charged centers are protected in the corruption and the net charge is invariant under every edit, so charge is preserved but not designed (Section 11). *Aromaticity.* States are Kekulé; “aromatic” is a perception on the round-tripped molecule, so `ring_system_restate` realizes aromatization and de-aromatization as validity-checked integer bond-order layouts rather than edits of a stored aromatic bond class.

C ALGORITHMS

We collect the three algorithms referenced in the main text. Algorithm 1 trains the universal edit prior by generator matching; Algorithm 2 performs one goal-conditioned controlled steer; Algorithm 3

Table 2: Production operator families. All families preserve connectivity, per-atom valence, and net charge (Proposition 1); implicit hydrogens re-derive from each touched atom’s valence class.

Family	Operands / applicability	Successor invariant & inverse	Training source
atom_insert	peripheral site + element/valence class + incident bond order	adds one heavy atom ($\leq N$); inverse = atom_delete	corruption; MMP graft-in
atom_delete	peripheral (leaf) atom whose removal keeps connectivity	removes one heavy atom; inverse = atom_insert	corruption; MMP peel
atom_restate	non-ring atom; new valence class feasible at its heavy-bond sum	element fixed, valence class changed, H re-derived; self-inverse family	corruption (bioisostere)
bond_reorder	non-ring (acyclic) bond; new integer order	bond order changed, endpoints’ H re-derived; self-inverse family	corruption
bond_reroute (Graft)	pendant subtree across a single bridging bond; target attachment site	subtree relocated, ring core fixed; scored at canonical successor group; inverse = reroute back	corruption; shared-scaffold
ring_system_grow	saturated chain match; typed ring template (mono/fused/bridged/spiro/aromatic/sat.)	adds a whole ring system with per-atom identity; inverse = ring_system_delete	retained from de novo prior
ring_system_delete	a supported ring system; decoration-preserving open	opens a ring keeping each atom’s element (Kekulé); inverse = ring_system_grow on the matching scaffold	corruption (ring-open, trim direction)
ring_system_restate	a ring system; aromaticity-changing restate (both directions)	changes perceived aromaticity via a validity-checked bond-order relayout; self-inverse family	corruption (aromatize $\leftrightarrow$ de-aromatize)

computes the exact controller on an enumerable state space. The Pareto-exploration loop repeatedly invokes Algorithm 2 with region-selected goals.

C.1 CONTROLLER DESIGN AND ITS EVALUATION AS AN APPROXIMATION

The amortized controller $h_\phi(b, x; x_{\text{src}}, z, m)$ approximates the finite-horizon value $h_{K-b}(x) = \mathbb{E}[g_z(X_K) \mid X_{b\text{-to-go}} = x]$ of Theorem 3; no separate control objective is introduced. Two estimators of this value target are consistent with the theory and are the design space we finalize empirically (and report as Appendix F keys): a rollout Monte-Carlo target, in which base-chain rollouts from x under the remaining budget estimate the expected desirability directly; and a bootstrapped backward-recursion target, in which $h_\phi(b, x)$ regresses onto $\sum_y P_\theta(y \mid x) h_\phi(b-1, y)$ with $h_\phi(0, \cdot) = g_z$. The parameterization conditions on the remaining budget b , source-relative features of x_{src} (similarity, protected-mask indicators), and an encoding of the goal z ; it shares the state encoder with the base model. Because the target is a value, not a policy, and because the controlled kernel (6) is self-normalized over the base support, an inaccurate h_ϕ cannot violate validity (it can only misallocate probability). We therefore evaluate h_ϕ as an approximation, reporting: (i) value calibration against the

Algorithm 1 Universal edit-prior training (generator matching)

Require: layered trace corpus (MMP / shared-scaffold / corruption), hierarchical sampler, model $(\Lambda_\theta, p_\theta)_\theta$

- 1: **while** not converged **do**
- 2: draw a layer by its mixture weight; draw a curriculum bin (measured path-length tercile); draw a balanced example under the per-scaffold / per-transformation / per-family caps
- 3: take a supervised committed transition (x, a^*, r) from the trace (both directions taught)
- 4: enumerate the factorized legal fiber at x ; verify a^* is in-fiber (else the loss is non-finite by design)
- 5: compute $\Lambda_\theta(x), p_\theta(a^* | x)$ (Graft scored at its canonical-successor group)
- 6: take a gradient step on the Poisson–Bregman loss (3)
- 7: **end while**
- 8: **return** θ

Algorithm 2 Goal-conditioned controlled editing (one steer)

Require: source x_{src} , budget K , base kernel P_θ , controller h_ϕ , goal z , protected mask m , objectives F , archive $\mathcal{A}$

- 1: $x \leftarrow x_{\text{src}}$; record $F(x)$
- 2: **for** $b = K, \dots, 1$ **do**
- 3: enumerate $\mathcal{A}(x)$ and aggregate to the canonical successor kernel $P_\theta(\cdot | x)$ {Proposition 2}
- 4: restrict support to marks respecting m and any active pathwise constraint {Corollary 5}
- 5: **if** support is empty **then**
- 6: **break** {terminal; every prefix is a valid molecule}
- 7: **end if**
- 8: form $P_b^\phi(x, \cdot)$ by (6) using $h_\phi(b-1, \cdot; x_{\text{src}}, z, m)$
- 9: sample $y \sim P_b^\phi(x, \cdot)$; commit $x \leftarrow y$; evaluate and record $F(x)$ { x is a complete molecule}
- 10: update $\mathcal{A}$ with x if nondominated and feasible
- 11: **end for**
- 12: **return** trajectory and updated archive $\mathcal{A}$

exact h on the enumerable benchmark; (ii) the backward-equation residual $\|h_\phi(b, \cdot) - P_\theta h_\phi(b-1, \cdot)\|$; and (iii) control quality (total variation to the exact terminal law) versus the exact controller. Where an unbiased terminal law is required despite residual value error, the optional twisted-SMC corrector of Section 6 is applied and its effective sample size reported. This section states the design honestly: the exact controller is realized and verified, and the amortized controller is trained and measured against it rather than assumed exact.

D DATA AND TRAINING

D.1 CORPUS AND SCOPE

The production corpus is the 500,000-molecule GuacaMol subset (standardized ChEMBL). Membership is decided by a single versioned *broad-organic* scope (broad_organic_v1): a molecule is in scope iff it is connected, has at most 40 heavy atoms, every element is in the active organic vocabulary (C, N, O, F, S, P, Cl, Br, I, B), and every atom’s exact (element, valence, charge) is a representable class under the same derivation the model’s heads use. Charged molecular states are *retained* (*retainrepresentablecharges*; *charge – preservingedits*); the scope decides membership only and never authorizes a charge-changing edit. The representation drops isotopes and stereochemistry and treats radicals or non-representable valences as out of scope. The 40-heavy-atom bound is the base model’s own state-space bound—the edit prior warm-starts from that architecture, and every intended benchmark lead lies within it—so it is part of the state-space *definition*, not a preprocessing cutoff. Under this scope 93.3% of the subset (466,483 molecules) is eligible, versus 48.0% (239,867) under the four-element (C/N/O/F) restriction it replaces: the broad vocabulary nearly doubles coverage while the atom bound retains most of it. The 33,484 rejected molecules are 32,171 above the 40-atom bound, 1,112 unsupported-element, and 201 unparseable; a further 33 accepted molecules are canonical duplicates removed post-standardization, so the 500,000 nonempty source lines reconcile exactly as eligible + duplicates + rejected. Every acceptance and rejection is counted

Algorithm 3 Exact enumerable controller (ground truth)

Require: source x_0 , budget K , base kernel P_θ , desirability g

- 1: enumerate the reachable set S from x_0 within K steps (breadth-first over the legal fiber); build the row-stochastic kernel P over S
 - 2: $h_K(x) \leftarrow g(x)$ for all $x \in S$
 - 3: **for** $k = K - 1, \dots, 0$ **do**
 - 4: $h_k(x) \leftarrow \sum_y P(x, y) h_{k+1}(y)$ for all $x \in S$
 - 5: **end for**
 - 6: steer with $P_k^g(x, y) = P(x, y) h_{k+1}(y) / h_k(x)$ where $h_k(x) > 0$
 - 7: **return** the exact controlled kernels $\{P_k^g\}$ and terminal tilt (5)
-

with an explicit reason, the scope carries a deterministic hash (3721d69851110fdd) recorded in every manifest and checkpoint, and mining/training use the TRAIN partition only, so val/test never leak.

D.2 FROM CORPUS TO SAMPLER

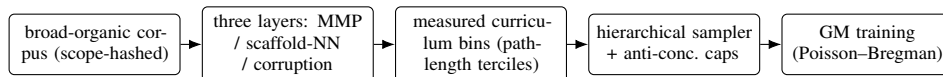


Figure 2: Corpus-to-sampler flow. Each stage records a version hash into the data manifest so the corpus cannot silently drift from the code that produced it.

D.3 TRACE LAYERS, COMPILER, AND REPLAY CHECKS

Compiled MMP analogue paths. One-cut matched molecular pairs (constant core; small variable fragment; bounded atom-count difference; no ring opened across the variable region) are compiled to executable rewrite programs using mmpdb’s constant-fragment and labelled-attachment mapping (not an unconstrained MCS): preserve the core, delete the source variable fragment leaf-to-root, retype and reorder as needed, insert the target fragment attachment-outward, replay through the *real* executor, and check endpoint graph isomorphism, in both directions. *Shared-scaffold analogue paths* connect each molecule to its k nearest analogue neighbours within a Bemis–Murcko group (sparse nearest neighbours, never all-pairs cliques). *Corruption paths* walk a real molecule a few reversible legal edits to a nearby valid source, taught both directions, and are the only layer that exercises the restate, bond-order, ring, and graft families at pilot scale. Every compiled path is verified for all-intermediate validity and connectedness, exact endpoint isomorphism, forward and inverse replay, repeated- and immediately-reversed-edit rates, protected-core invariance, operator-family usage, and the round-trip $A \rightarrow B \rightarrow A$; a path that reaches the right canonical SMILES but breaks the attachment mapping or needs gratuitous cycles is rejected. The locked production pool compiles 363,456 directed edit records from 216,183 one-cut constant-core groups (178,797 forward / 184,659 reverse; the compiled pool is exactly direction-balanced and the residual imbalance is entirely the per-source cap, which keys on each record’s source molecule), with curriculum terciles at path lengths 5 / 9 / 13 and the pool, operator-registry (34234a88b8cc642a), ring-catalog (639ff6078c32d43c), and standardization (a8d574cc91c89cb3) hashes pinned in the manifest.

D.4 ANTI-CONCENTRATION SAFEGUARDS AND CURRICULUM

Each safeguard has a measured signal and a gate: identity traces are dropped (identity fraction ≈ 0); a per-scaffold cap bounds Bemis–Murcko-scaffold Gini and top- k share; a per-transformation cap and de-duplication bound transformation top- k share; and every enabled family is floored to a minimum supervised-trace count (rare families are covered by oversampling corruption instances that exercise them). Curriculum bins are the *measured* terciles of compiled path length (micro / lead-opt / scaffold), refined by ring complexity and attachment count—never preset. The hierarchical sampler draws a layer by its (proposed) mixture weight, then a curriculum bin, then a balanced example under the caps.

D.5 PILOT CHARACTERIZATION (WIRING FIXTURE, NOT THE PRODUCTION MIXTURE)

At pilot scale (700 corruption traces from real→corrupt and corrupt→real, and 98 analogue traces from 49 one-cut MMP pairs): the corruption layer has 6 active families, scaffold Gini 0.14, aromatic-ring change on 47% of traces, source–target median Tanimoto 0.40, and identity fraction 0.29%; the one-cut analogue layer contributes only insert/delete by construction, which is why a corruption-only or analogue-only corpus is wrong. This pilot pool is a wiring fixture—its scaffold concentration is too high and its analogue count too small to be the production analogue layer—and its percentages are *not* the production mixture; the proposed starting shares ($\sim 50\%/\sim 35\%/\sim 15\%$) are an ablation target finalized only from the at-scale statistics. A versioned data manifest records the partition, standardization and operator-registry hashes, compiler versions, split identifiers, trace counts, and curriculum-bin definitions.

D.6 TRAINING

The edit prior is warm-started from the de novo carbon-tree base checkpoint: shape-compatible body and embedding parameters transfer, and the wider organic (element, valence) heads (width 15, versus 4 for the four-element base) retain the first four transferred rows and freshly initialize the rest, so a four-element checkpoint loads without a width crash and a de novo checkpoint reloads byte-identically. We state the warm-start honestly: it carries the de novo ring-growth vocabulary (which the edit layers do not teach) and is a starting point, not the trained edit prior. Save/reload is exact—a reloaded checkpoint reproduces the sampled mark, hazard, and a fixed-seed trajectory, and a checkpoint missing required metadata fails loudly. The optimizer, schedule, batch size, loss normalization, precision, hardware, step budget, checkpoint selection (held-out validation only, never the test set), and seeds are recorded with each run and released with the code. “Universal” means one edit prior shared across sources and objectives within the declared vocabulary—not all of chemical space.

E IMPLEMENTATION VALIDATION

This appendix reports *implementation-validation* evidence: checks that the code implements the stated mathematics. These are not headline empirical results (which are the RQ studies of Section 9, carried as registry placeholders), and we are careful not to present a passing test as a scientific finding.

E.1 ENUMERABLE-BENCHMARK CONSTRUCTION AND DOOB RESIDUALS

The enumerable chain is built by breadth-first expansion of the real legal fiber from methane, capping successors at 4 heavy atoms, yielding a 572-state row-stochastic embedded kernel over canonical molecular states. On this chain the finite-horizon Doob transform is verified against its closed form: the controlled terminal law matches the analytic tilt to total variation 8×10^{-17} (Theorem 3); dynamic-retargeting continuation matches the g' -tilt from the switch state to 9×10^{-17} (Corollary 4); and an early-stop marginal is confirmed *not* to equal the terminal tilt ($TV > 2 \times 10^{-2}$ two steps early), so exactness is budget-specific. No sampled controlled transition leaves the base support.

E.2 KERNEL, MEASURE, AND QUOTIENT

The successor-sampling kernel of the canonical-successor quotient sampler equals the raw mark sampler’s induced kernel to $< 10^{-5}$ on tree, symmetric-tree, self-graft, cyclic, and post-ring-opening states (Proposition 2); the mark law is a proper distribution ($\sum_a p_\theta(a | x) = 1$ to $\sim 10^{-7}$ on every molecule tested), Graft group mass equals the sum of its successor-group rates, and Monte-Carlo successor frequencies agree within the null band with no out-of-support draws.

E.3 OBJECTIVE, GRADIENTS, AND TINY-CORPUS CONVERGENCE

On a tiny verified corpus over distinct states (no conflicting targets), the Poisson–Bregman loss descends from 5.23 to 1.14, within 0.14 of the per-example lower bound 1.0, and every production family’s selected-mark probability rises—so each enabled head receives a finite, non-suppressive gradient from a positive teacher, and the model can overfit the verified corpus. This is a wiring check, not a quality claim.

E.4 CLOSURE, CHARGE, SAVE/RELOAD, HELD-OUT ROLLOUTS

A random legal-edit fuzz produces valid, connected successors across families (Proposition 1); charged, drug-like leads round-trip with net charge and charged centers preserved (no edit changes a formal charge); save/reload is exact and reconstructs the organic vocabulary and editing capabilities from checkpoint metadata (a de novo checkpoint reloads byte-identically; a checkpoint missing required metadata fails loudly); and over held-out drug-like leads at budgets 1–8, rollouts start at the lead (never a carbon-tree seed), recompute the legal set per state, commit only executor-accepted edits, and maintain 100% all-intermediate validity and connectivity. The degeneracy/reversal diagnostics are measured for plumbing; their *quality* reading requires the trained model.

E.5 TEST SUITE AND CLEAN-LINEAGE PROVENANCE

The committed suite passes on a pristine worktree (recorded gates: 453 at the audited commit and 463 at the Stage-6 commit; the count grows with each added check). The evidence lineage for this manuscript is commit ee5a6ca on the clean lineage tagged stage6-preflight-baseline. Pre-fix editing outputs (which sampled the de novo vocabulary for an edit checkpoint) are treated as stale and are never cited.

E.6 LEGACY-SAMPLER CORRECTNESS DIAGNOSTIC (NOT A COMPETITOR)

A prior sampler variant normalized the Graft family over the raw (self-graft-inclusive) partition rather than the quotient partition, giving a graft family rate that disagreed with training (e.g. 0.097 vs 0.136 on octane). This is retained only as a named historical ablation to *diagnose* the quotient convention; it is not a controller and is never run as a baseline in Section 9, because a known-incorrect sampler is not a fair scientific competitor.

F EXTENDED EXPERIMENTAL PROTOCOL

This appendix gives the full protocol behind Section 9: metric definitions stated before any result, per-baseline documentation, and the figure/table scaffolds. All empirical cells are result-registry placeholders.

F.1 RQ1 PANELS (ENUMERABLE BENCHMARK)

Each conditioning task is evaluated by total variation between a method’s controlled terminal law and the exact target: exact Doob (xxx, with maximum per-state error xxx), the learned controller (xxx), one-step greedy (xxx), endpoint reranking (xxx), and dynamic retargeting (xxx); the target-event-probability error under indicator conditioning is xxx, and support violations are xxx (zero by construction).

FIGURE PENDING

renders from a machine-readable artifact when the run completes

Panels: (a) unconditioned vs desired-tilted vs exact-Doob vs greedy endpoint histograms on the enumerable chain; (b) TV to the exact law per method; (c) h_ϕ calibration vs exact h ; (d) backward-equation residual vs training. Rendered from the enumerable-benchmark artifact.

Figure 3: **Exactness on the enumerable chemical state space (RQ1).** Exact Doob is expected to match the analytic target to numerical precision while reward-local controllers attain high reward at a different distribution; the figure is populated from the enumerable-benchmark artifact when the run completes.

F.2 ADAPTATION REGRET (DEFINED BEFORE RESULTS)

For a preference switch to z' at trajectory fraction τ/K , let $U_{z'}$ be the new utility and $U_{z'}^*(x_\tau, b)$ the best achievable expected utility from state x_τ with remaining budget $b = K - \tau$ under the base process. The *adaptation regret* of a method that reaches terminal utility $U_{z'}(X_K)$ is $R = U_{z'}^*(x_\tau, b) - \mathbb{E}[U_{z'}(X_K)]$, estimated with the exact controller as the reference on enumerable slices

and with a matched-budget oracle-best on real leads; lower is better. Steps-to-region counts committed edits until $F(X_k) \in \mathcal{B}_{z'}$, and retention is the fraction of the pre-switch objective’s gain preserved at the terminal state. These are fixed here so RQ3 reports them without post-hoc definition.

F.3 PER-BASELINE DOCUMENTATION

Same-base controllers (all steer the identical canonical-successor kernel P_θ): *unguided* samples P_θ ; *endpoint+rerank* samples P_θ and reranks terminal candidates by the objective; *greedy* ΔF picks the successor of maximal immediate objective gain; *local Boltzmann* reweights P_θ by $\exp(\beta\Delta F)$; *MOG-DFM-style* applies a rank-directional score with an adaptive hypercone filter over P_θ ’s successors; *SMC/Feynman-Kac* runs a twisted-SMC population with resampling; *learned Doob* uses (6). *External optimizers* (contextual; Table 3): GraphGA (graph genetic algorithm), MARS (annealed fragment-edit MCMC), GenMol (masked discrete-diffusion generalist), and RetMol (retrieval-controlled generation); each is run at a matched oracle budget with its released configuration, and where possible additionally over our legal proposal set to separate the control law from the edit vocabulary. Any asymmetric tuning is disclosed with the result.

F.4 HYPERVOLUME HYGIENE

Objective bounds and the reference point are fixed before evaluation and published; the archive is capped per method; hypervolume is reported against oracle calls (HV-AUC), not only at the final budget; and hypervolume is paired with IGD+ and coverage so a single indicator does not carry the claim. All oracle evaluations—including intermediates, rejected, invalid, reranked, and SMC particles—are counted (Section 10).

F.5 SECONDARY: DE NOVO GENERATION

De novo generation from the target-independent carbon-tree prior is evaluated on all-step validity (xxx), uniqueness, novelty, Fréchet ChemNet Distance (xxx), and property divergence, as evidence that the operators form a genuine generative process. It is secondary; the paper’s claims do not depend on carbon-tree intermediates being useful lead-optimization states.

Table 3: **RQ2 external optimizers (contextual)**. These methods differ from ours in scale and representation, so they contextualize rather than isolate the control law; where possible each is additionally run on our legal proposal set (Section 10). Same conventions and reference point as Table 1; the Diversity column (internal diversity of the nondominated set) is included here for completeness. Placeholders are filled from the registry when the runs complete.

Method	nHV $\uparrow$	HV-AUC $\uparrow$	IGD+ $\downarrow$	#Feas $\uparrow$	Src-sim $\uparrow$	Div $\uparrow$	Orc-eff $\uparrow$
GraphGA	xxx	xxx	xxx	xxx	xxx	xxx	xxx
MARS	xxx	xxx	xxx	xxx	xxx	xxx	xxx
GenMol	xxx	xxx	xxx	xxx	xxx	xxx	xxx
RetMol	xxx	xxx	xxx	xxx	xxx	xxx	xxx
<i>Same-base controllers (Table 1) additionally with the Diversity column:</i>							
Learned Doob (ours)	xxx	xxx	xxx	xxx	xxx	xxx	xxx

G LARGE LANGUAGE MODEL USE

We disclose the use of large language models (LLMs) in preparing this work, factually and specifically. An LLM is not an author and made no scientific decisions. LLM assistance was used for: (i) drafting and copy-editing prose from author-provided technical content, notation, and results, which the authors then verified line by line against the implementation and the mathematics; (ii) assisting a literature search whose every citation was subsequently verified by the authors against a primary source (arXiv, OpenReview, or a publisher DOI), with venue and author details corrected where the model erred; and (iii) routine code and tooling assistance (for example, the result-placeholder build system) under author review and the project’s test suite. All theorem statements, proofs, experimental designs, claims, and their evidentiary status were determined and checked by the authors; the LLM did not generate any empirical number, and the result-registry system exists precisely so that no value enters the manuscript without a machine-readable artifact behind it. The authors take full responsibility for the entire content, including any remaining errors.

FIGURE PENDING

renders from a machine-readable artifact when the run completes
One lead x_{src} ; a shared controlled prefix; then a fan of preference-conditioned valid branches
(one per goal z), each ending in a committed molecule. Molecules rendered from a real run;
branch selection, if any, disclosed here.

Figure 4: **Dynamic preference steering: the Pareto fan (RQ3).** From a single source lead, a shared controlled prefix fans into several preference-conditioned trajectories via exact continuation (Corollary 4); each branch is a distinct goal descriptor z applied to the remaining budget, and every committed state along every branch is a complete, valid molecule. The panel is populated from one real dynamic-steering run over a single lead across 8–16 preferences; molecules shown are actual trajectory states (no cherry-picked composites), and any down-selection among branches for legibility is stated in this caption when the figure is rendered. No numerical results are asserted here until the run completes.

H ANONYMOUS-RELEASE CHECKLIST

A code and data release accompanying this paper will be scrubbed for double-blind review against the following checklist before posting; each item is a required pre-release check. *Identity*: remove author and affiliation names, acknowledgements, and grant numbers; remove any user, account, or cluster names. *Paths and infrastructure*: rewrite absolute home-directory paths, remove cloud-profile and compute-account names, and strip internal hostnames and repository remotes. *Self-citation*: phrase all references to prior work by the same authors in the third person and remove links that would reveal identity through a personal or lab domain. *Artifacts*: scrub PDF metadata (author, title, producer), strip identifying strings from supplementary archives and notebooks, and remove identifying comments and environment files from the code. *Links*: replace tracking or identity-revealing URLs with anonymized mirrors. *Terminology*: the manuscript uses only public method terminology (base prior, edit prior, canonical successor kernel, controlled editing process, source-conditioned controller); internal engineering codenames are confined to the authors’ private reproducibility notes and never appear in the released paper or code comments. The released bundle will include the result registry, the plotting artifacts backing every figure and table, the enumerable-benchmark harness, and the data manifest with its scope and operator-registry hashes, so that every claim in the paper maps to a runnable check.